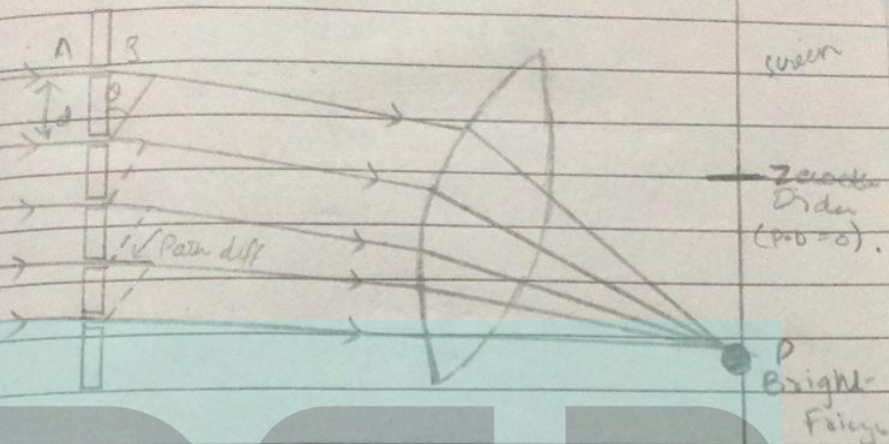


DIFFRACTION GRATING

DEFINITION:-

A piece of glass or plastic



Consider $\triangle AOB$:-

$$\sin \theta = \frac{P}{H}$$

$$\sin \theta = \frac{AB}{OA}$$

$$\sin \theta = \frac{P.D}{d}$$

$$P.D = d \sin \theta$$

$$P.D = m\lambda$$

$$m\lambda = d \sin \theta$$

→ Condition for Bright Fringe.

$$m = 0, 1, 2, 3, \dots$$

$$\therefore 0 = \text{Zeroth Order.}$$

Teacher's Signature

Chap 1-13

Date _____

Numericals

Q11 Data:-

$$\lambda = 6900 \text{ \AA}$$

$$= 6900 \times 10^{-1}$$

$$L = 2.30 \text{ m}$$

$$d = ?$$

$$\Delta x = 1.86 \text{ cm} \approx 0.018 \text{ m}.$$

Sol.

$$\Delta x = \frac{L\lambda}{d}$$

$$d = \frac{L\lambda}{\Delta x}$$

$$=$$

Q12

Ans:-

$$d = 25.8 \text{ nm} \rightarrow 25.8 \times 10^{-6} \text{ m}$$

$$m = 92$$

$$\lambda = ?$$

$$d = \frac{m\lambda}{2}$$

$$\lambda = \frac{2d}{m}$$

$$\lambda = \frac{2nt}{m}$$

$$= \frac{2 \times 1.5 \times 910 \times 10^9}{m}$$

$$\lambda = \frac{2730}{m}$$

Now, for visible light :-

$$\lambda = 400 \text{ nm} \sim 700 \text{ nm}$$

$$\rightarrow \underline{m = 4 :-}$$

$$\lambda_4 = 682.5 \text{ nm}$$

$$\rightarrow \underline{m = 8 :-}$$

$$\lambda_8 = 546 \text{ nm}$$

$$\rightarrow \underline{m = 6 :-}$$

$$\lambda_6 = 455 \text{ nm}$$

Part b:-

for constructive interference :-

$$2nt = \left(m + \frac{1}{2}\right) \lambda$$

$$+ \frac{1}{2} = 0.5$$

$$2nt = (m + 0.5)\lambda$$

$$\lambda = \frac{2nt}{m + 0.5}$$

$$\lambda = \frac{273 \text{ nm}}{m + 0.5}$$

$$\rightarrow \underline{m = 4}$$

$$\lambda_4 = 606.6 \text{ nm}$$

$$\rightarrow \underline{m = 5:-}$$

$$\lambda_5 = 496.3 \text{ nm}$$

$$\rightarrow \underline{m = 6:-}$$

$$\lambda_6 = 420 \text{ nm}$$

Q12 Dates:-

$$a = 0.020 \text{ nm} \rightarrow 0.02 \times 10^{-9}$$

$$L = 1.20 \text{ m}$$

$$\lambda = 430 \text{ nm} \rightarrow 430 \times 10^{-9}$$

$$\Delta y = ?$$

Sol:

$$\Delta y = \frac{2N}{a}$$

$$= \frac{2 \times 430 \times 10^{-9} \times 1.20}{0.02 \times 10^{-1}}$$

$$= 2 \times 430 \times 10^{-9} \times 1.20$$

$$= 251600 \text{ m}$$

$$\Delta y = 5.16 \text{ cm}$$

Q 13:-

Data:-

$$N = 8000 \text{ lines}$$

$$L = 2.54 \text{ cm}$$

$$= 2.54 \times 10^{-3} \text{ m}$$

$$m = 3$$

$$\theta = ?$$

$$\lambda = 546 \text{ nm}$$

$$= 546 \times 10^{-9}$$

grating
element.

$$d \sin \theta = m \lambda$$

$$\theta = \sin^{-1} \left(\frac{m \lambda}{d} \right)$$

$$d = \frac{L}{N}$$

$$= \frac{0.04 \times 10^{-3}}{2000}$$

$$= 3.175 \times 10^{-6}$$

$$\theta = \sin^{-1} \left(\frac{m\lambda}{d} \right)$$

$$= \sin^{-1} \left(\frac{3 \times 546 \times 10^{-9}}{3.175 \times 10^{-6}} \right)$$

$$= \sin^{-1} \left(\frac{0.518}{1} \right)$$

$$\theta = 31^\circ$$

$$5.60 \times 10^{-9}$$

Q₂ → Reasoning

Q₄ Data:-

$$\lambda = 400 \text{ nm}$$

$$\Delta t = t_6 - t_3 = ?$$

$$R = 5 \text{ m}$$

b) ~~3~~ Bright = ?

a) $\Delta t = ?$

$$2mr = \left(m + \frac{1}{2}\right) \lambda$$

for air, $m = 1$

$$2t = \left(m + \frac{1}{2}\right) \lambda$$

For 3rd Bright Fringe:- ($m = 2$)

$$2t_3 = \left(2 + \frac{1}{2}\right) \lambda$$

$$2t_3 = \frac{0}{2} \lambda$$

$$t_3 = \frac{0}{4} \lambda$$

For 6th Bright Fringe ($m=8$).

$$2t_6 = \left(8 + \frac{1}{2}\right) \lambda$$

$$2t_6 = \frac{11}{2} \lambda$$

$$t_6 = \frac{11}{4} \lambda$$

$$\underline{\Delta t =}$$

$$= t_6 - t_3$$

$$= \frac{11}{4} \lambda - \frac{0}{4} \lambda$$

$$= \frac{11}{4} \lambda$$

$$= \frac{11}{4} (400 \text{ nm})$$

$$\Delta t = 1100 \text{ nm}$$

b) 83B 1-

$$\delta = \sqrt{\left(\frac{m+1}{2}\right) \lambda R}$$

$$= \sqrt{\left(\frac{2+1}{2}\right) \lambda R}$$

$$= \sqrt{\frac{3}{2} \lambda R}$$

$$= \sqrt{\frac{3}{2} (8) (400 \times 10^9)}$$

$$= 2.236 \times 10^{-3} \text{ m}$$

Q 51-

Data:-

$$n = 1.50$$

$$t = 9.10 \text{ nm}$$

a) $\lambda_{\text{miss}} = ?$

b) $\lambda_{\text{V. noyes / visible / appeared}} = ?$

Sol:-

a) For destructive interference

$$2nt = m\lambda$$

Newton's Formula For Speed Of Sound

Assumptions:-

- Isothermal process (Temp. constant).
- Sound propagation is a slow process. (Enough time to transfer heat).

Mathematical Working:-

→ Since sound waves are longitudinal waves.

$$\Rightarrow V = \frac{\sqrt{E}}{\sqrt{\rho}} \rightarrow \text{Elasticity}$$

⇒ Using Boyle's law:-

$$P \propto \frac{1}{V}$$

$$P = (\text{const}) \frac{1}{V}$$

$$PV = (\text{const})$$

$$P_1 V_1 = P_2 V_2$$

$$\therefore P_1 = P$$

$$V_1 = V$$

$$\Delta P = P_2 - P_1$$

$$\therefore P_2 = \Delta P + P_1$$

$$V_2 = V_1 - \Delta V$$

$$P_2 V_2 = (P_1 + \Delta P)(V_1 - \Delta V)$$

$$P_2 V_2 = P_1 V_1 + P_1 \Delta V + \Delta P V_1 - \Delta P \Delta V$$

$\rightarrow$ In sound propagation $\Delta P \Delta V$ is very small, hence negligible.

$$0 = -P_1 \Delta V + \Delta P V_1$$

$$P_1 \Delta V = \Delta P V_1$$

$$P = \frac{\Delta P}{\frac{\Delta V}{V_1}}$$

$$\Rightarrow P = \frac{\Delta P}{\left(\frac{\Delta V}{V}\right)}$$

Here:-

$\Delta P = \text{Stress}$

$\frac{\Delta V}{V} = \text{Strain}$

$$P = \frac{\text{Stress}}{\text{Strain}}$$

$$P = E$$

* LALACE CORRECTION :-

Date _____

★ Assumption :-

- Sound wave propagation is an Adiabatic Process (No heat transfer).
- Sound wave propagation is faster / rapid process (No enough time)

Mathematical Der. Working :-

• Modification in Boyle's Law :-

$$PV^\gamma = \text{const}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$PV^\gamma = (P + \Delta P)(V - \Delta V)^\gamma$$

$$PV^\gamma = (P + \Delta P) \left[V \left(1 - \frac{\Delta V}{V} \right) \right]^\gamma$$

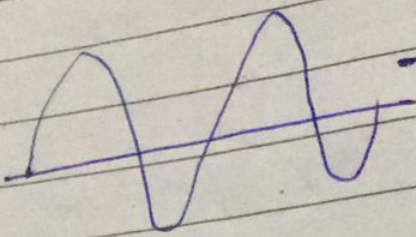
$$PV^\gamma = (P + \Delta P) V^\gamma \left(1 - \frac{\Delta V}{V} \right)^\gamma$$

$$\Rightarrow P = (P + \Delta P) \left(1 - \frac{\gamma \Delta V}{V} \right) \rightarrow (1 - x)^\gamma \approx 1 - \gamma x$$

$$P = P - \frac{\gamma P \Delta V}{V} + \Delta P - \frac{\gamma \Delta V \Delta P}{V} \quad \text{negligible}$$

$$0 = -\frac{\gamma P \Delta V}{V} + \Delta P$$

Date _____



→ Sinusoidal
Wave.

$$\frac{\gamma \Delta PV}{V} = \Delta P$$

$$\gamma P = \frac{\Delta PV}{\Delta V}$$

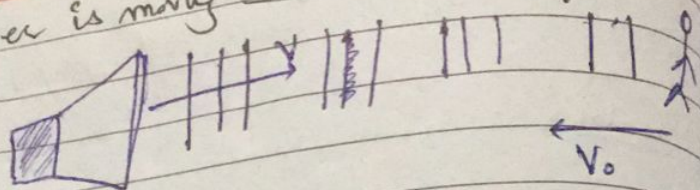
$$\gamma P = \frac{\Delta P}{\left(\frac{\Delta V}{V} \right)}$$

$$\gamma V = \sqrt{\frac{E}{\rho}}$$

Date _____

Teacher's Signature _____

a) Observer is moving towards stationary source.



$$V_s = 0$$

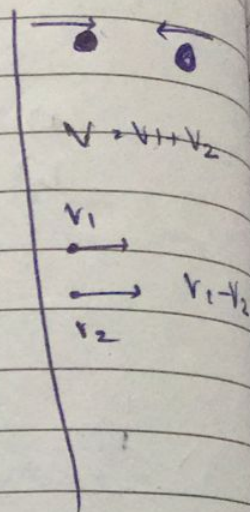
$$V = f\lambda$$

$$f = \frac{V}{\lambda}$$

$$\text{or } \lambda = \frac{V}{f}$$

$$f' = \frac{V'}{\lambda}$$

$$\therefore V' = V + V_o$$



$$f' = \frac{V + V_o}{\frac{V}{f}}$$

$$f' = \frac{V + V_o}{V} f$$

Hence -

$$f' > f$$

- High Pitch
- shrill sound

* DOPPLER'S EFFECT

$v_s \rightarrow$ Velocity of source

$v_o \rightarrow$ Velocity of observer

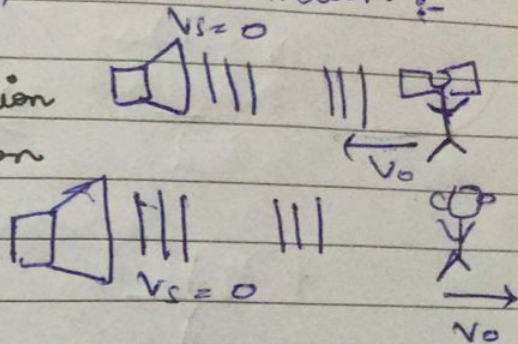
$\rightarrow$ The apparent change in freq of sound due to relative motion b/w source & listener/observer.

Cases :-

i) Source is stationary, observer in motion :-

a) Towards Motion

b) Away motion



ii) Observer is stationary, Source is in motion

a) Towards motion

b) Away motion

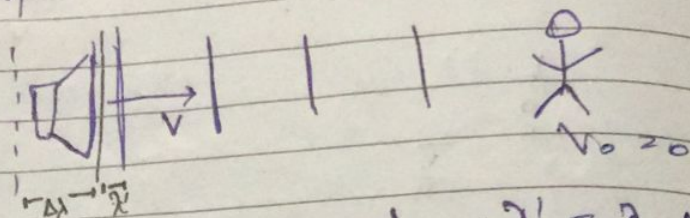
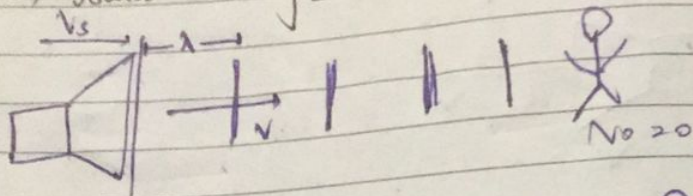
iii) Both are moving :-

a) Towards Motion

b) Away Motion

Teacher's Signature

c) Source is moving towards stationary observer :-



$$v = f \lambda$$

$$f = \frac{v}{\lambda}$$

$$\Rightarrow f' = \frac{v}{\lambda}$$

$$\therefore \lambda' = \lambda - \Delta \lambda$$

$$\lambda = \frac{v}{f} - \frac{v_s}{f}$$

$$\therefore v = f \lambda'$$

$$\lambda = \frac{v}{f}$$

$$\Delta \lambda = \frac{v_s}{f}$$

$$\lambda' = \frac{v - v_s}{f}$$

$$\Rightarrow f' = \frac{v}{\left(\frac{v - v_s}{f} \right)}$$

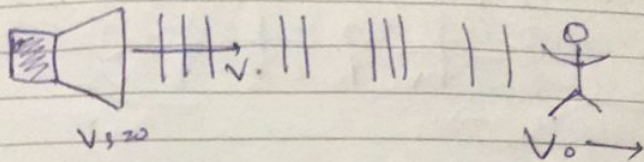
$$f' = \left(\frac{v}{v - v_s} \right) f$$

Hence,

$$f' > f$$

→ Shill sound.
→ High Pitch.

b) Observer is moving against ~~Source~~ Stationary source:-



$$V = f\lambda$$

$$f = \frac{V}{\lambda}$$

$$f' = \frac{V'}{\lambda}$$

$$\& \neq \quad \rightarrow V' = V - V_o$$

$$\therefore \lambda = \frac{V}{f}$$

$$f' = \frac{V - V_o}{\frac{V}{f}}$$

$$f' = f \left(\frac{V - V_o}{V} \right)$$

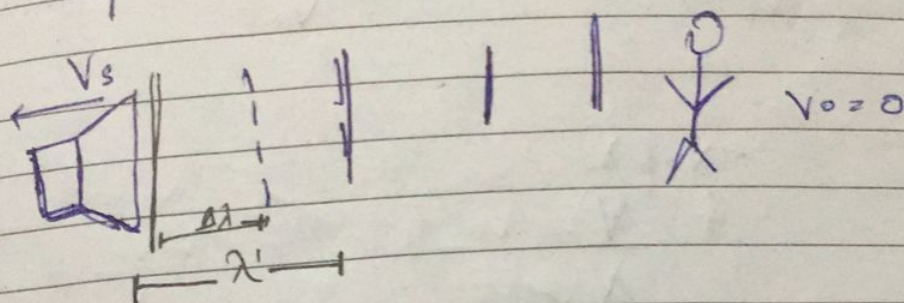
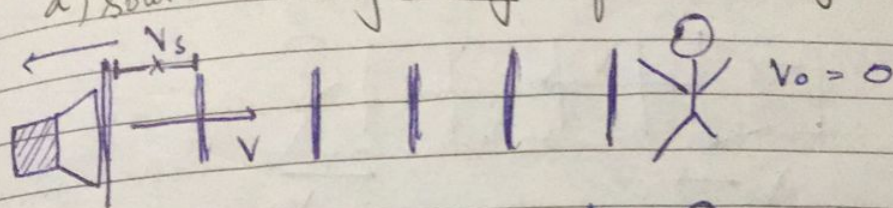
Hence:-

$$f' < f$$

• low pitch

• Grave sound.

d) Source is moving away from stationary observer



$$v = f \lambda$$

$$f = \frac{v}{\lambda}$$

$$f' = \frac{v}{\lambda'}$$

$$\left. \begin{aligned} \therefore \lambda' &= \lambda + \Delta \lambda \\ \lambda' &= \frac{v}{f} + \frac{v_s}{f} \\ \Rightarrow \lambda' &= \frac{v + v_s}{f} \end{aligned} \right\}$$

$$f' = \frac{v}{\left(\frac{v + v_s}{f} \right)}$$

$$f' = \left(\frac{v}{v + v_s} \right) f$$

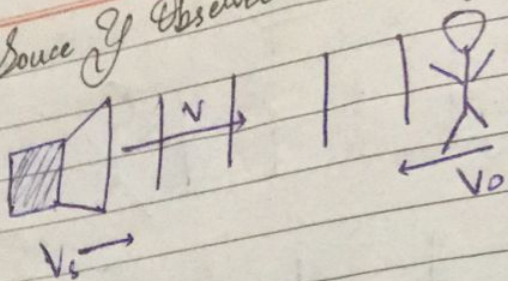
$$\boxed{f' < f}$$

Hence -

→ Grave

→ Low

c) Source & Observer are moving towards each other.



$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$\Rightarrow f' = \frac{v'}{\lambda'}$$

$$f' = \frac{v + v_o}{\left(\frac{v - v_s}{f}\right)}$$

$$f' = \left(\frac{v + v_o}{v - v_s}\right) f$$

$$f' > f$$

Hence:-

→ High Pitch

→ Shril sound.

* General Form for Doppler's Effect :-

★

$$f' = \left(\frac{V \pm V_o}{V \pm V_s} \right) f$$

Towards :-

Source stationary $V_s = 0$

$$f' = \left(\frac{V + V_o}{V + V_s} \right) f \quad \leftarrow f' = \left(\frac{V + V_o}{V} \right) f$$

Observer stationary $V_o = 0$

$$f' = \left(\frac{V + V_o}{V - V_s} \right) f$$

Ways :-

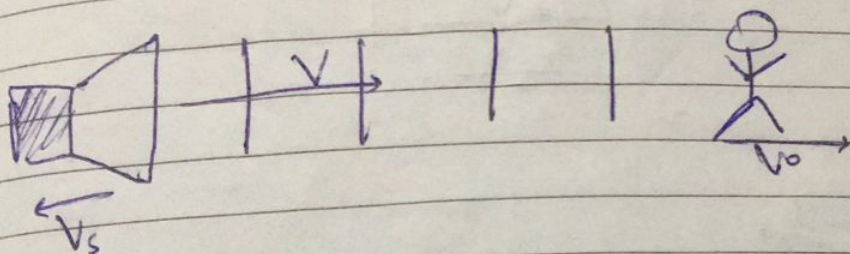
Source stationary $V_s = 0$

$$f' = \left(\frac{V - V_o}{V + V_s} \right) f \quad \leftarrow f' = \left(\frac{V - V_o}{V} \right) f$$

Observer stationary $V_o = 0$

$$f' = \left(\frac{V}{V + V_s} \right) f \quad \checkmark$$

f) Source & Observer are moving away from each other.



$$\because v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f' = \frac{v'}{\lambda'}$$

$$f' = \frac{v - v_o}{\left(\frac{v + v_s}{f} \right)}$$

$$f' = \left(\frac{v - v_o}{v + v_s} \right) f$$

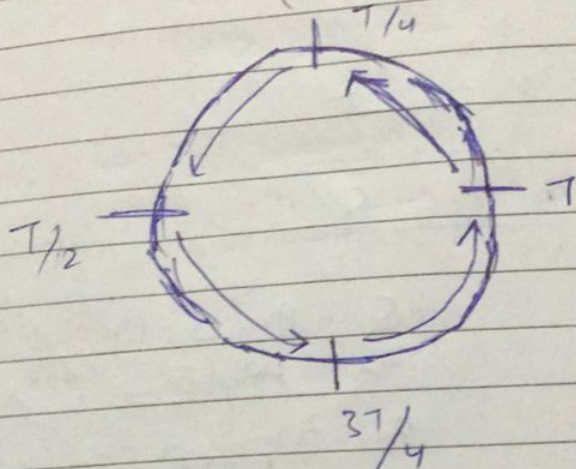
$$f' < f$$

→ For 1 complete rev:-

$$\theta = 2\pi$$

$$t = T$$

$$x = x_0 \cos\left(\frac{2\pi}{T} \cdot t\right)$$



① When $t = 0$

$$\underline{t = 0}$$

$$x = x_0 \cos\left(\frac{2\pi}{T} \cdot 0\right)$$

$$x = x_0 \cos(0)$$

$$\Rightarrow \cos 0 = 1$$

$$x = x_0(1)$$

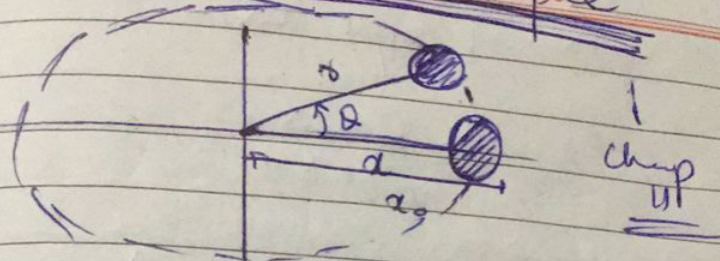
$$x = x_0$$

Max. displacement

No. _____

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Graphical Representation of SHM by Superposition Principle



$$\cos \theta = \frac{x}{r}$$

$$x = r \cos \theta$$

$$x = x_0 \cos \theta$$

θ = Angle (Phase),

x_0 = Amplitude (Max. displacement),

x = Displacement from mean position.

θ Tells instantaneous displacement-direction of " x ".

Now,

$$\theta = \omega t$$

$$x = x_0 \cos(\omega t)$$

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(2) when: -

$$t = \frac{T}{4}$$

$$\Rightarrow x = x_0 \cos\left(\frac{2\pi \cdot T}{T} \cdot \frac{T}{4}\right)$$

$$x = x_0 \cos\left(\frac{\pi}{2}\right)$$

$$\pi = 180^\circ$$

$$x = x_0 \cos\left(\frac{180}{2}\right)$$

$$x = x_0 \cos(90)$$

~~the~~

$$\cos(90) = 0$$

$$x = x_0(0)$$

$$\boxed{x = 0} \longrightarrow \text{Mean}$$

⑤ When $\rightarrow$

$$t = T$$

$$x = x_0 \cos\left(\frac{2\pi}{T} \cdot T\right)$$

$$x = x_0 \cos(2\pi)$$

$$x = x_0 \cos(360)$$

$$\cos(360) = 1$$

$$x = x_0(1)$$

$$\boxed{x = x_0}$$

③ When:-

$$t = \frac{T}{2}$$

$$\Rightarrow x = x_0 \cos \left(\frac{2\pi}{T} \cdot \frac{T}{2} \right)$$

$$x = x_0 \cos(180^\circ)$$

$$+ \cos(180^\circ) = -1$$

$$\Rightarrow x = x_0 \cos(180^\circ) = x_0(-1)$$

$$\boxed{x = -x_0} \rightarrow \text{opposite to mean.}$$

④ When:-

$$t = \frac{3T}{4}$$

$$\Rightarrow x = x_0 \cos \left(\frac{2\pi}{T} \cdot \frac{3T}{4} \right)$$

$$x = x_0 \cos \left(\frac{3\pi}{2} \right)$$

$$x = x_0 \cos(270^\circ)$$

$$? \cos(270^\circ) = 0$$

$$x = x_0(0)$$

$$\boxed{x = 0} \rightarrow \text{mean.}$$

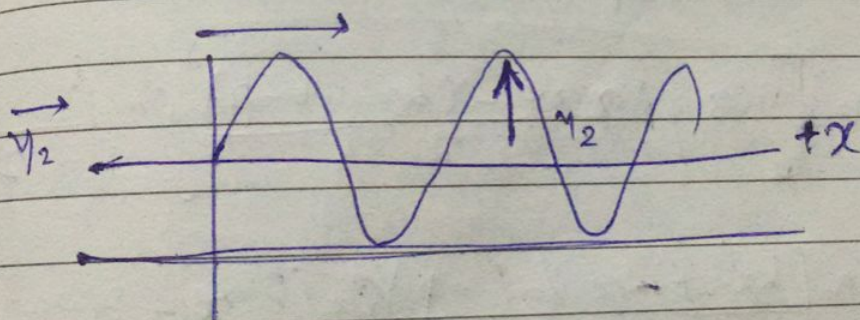
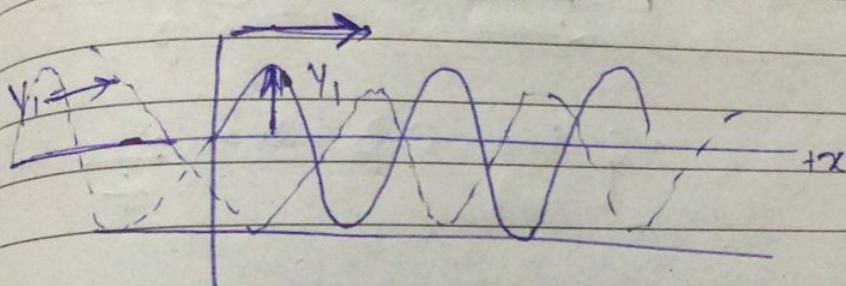
Mathematical Treatment Of Superposition

Date _____

Let y_1 and y_2 representing the individual displacement of two waves which differ in phase (ϕ) by a phase constant ϕ .
The eq. of wave can be written as:-

$$y_1 = A_0 \sin(kx - \omega t) \quad "A_0 \rightarrow \text{Amplitude.}$$

$$y_2 = A_0 \sin(kx - \omega t - \phi)$$



y = Instantaneous displacement.

A_0 = Amplitude.

k = Propagation const. or wave no.
 $= \frac{2\pi}{\lambda}$

x = Horizontal displacement.

ω = Angular velocity $\omega = \frac{2\pi}{T}$

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Superposition of Waves Date

Statement:

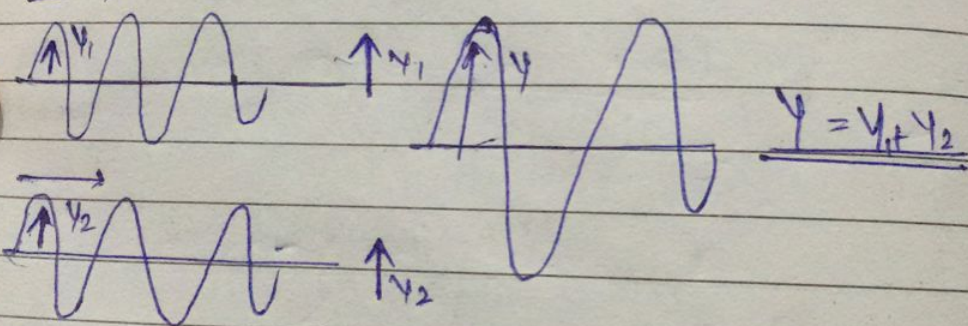
When 2 or more waves meet at one point in the same medium. The resultant amplitude at that point is algebraic sum of the amplitudes of individual waves.

$$Y = Y_1 + Y_2 + Y_3$$

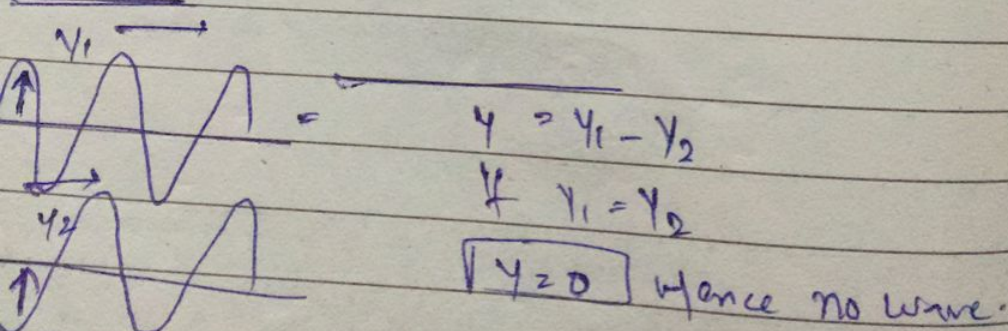
EFFECTS OF SUPERPOSITION OF WAVES

- 1) Interference of waves
- 2) Beats
- 3) Stationary waves.

Case 1



Case 2



Teacher's Signature

$t = \text{time}$ $\phi = \text{phase const.}$ wave no. $\rightarrow$ no. of waves per unit lengthApply principle of Superposition of waves

$$\rightarrow y = y_1 + y_2$$

$$y = \frac{1}{2} A_0 \sin \left(\frac{kx - \omega t}{2} \right) + A_0 \sin \left(\frac{kx - \omega t - \phi}{2} \right)$$

$$+ \frac{1}{2} \sin \alpha + \sin \beta = 2 \cos \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right)$$

$$y = A_0 \left[2 \cos \left(\frac{kx - \omega t - (kx - \omega t - \phi)}{2} \right) \right] \sin \left(\frac{2kx - \omega t - \phi}{2} \right)$$

$$= A_0 2 \cos \left(\frac{kx - \omega t - kx + \omega t + \phi}{2} \right) \sin \left(\frac{2kx - 2\omega t - \phi}{2} \right)$$

$$y = A_0 \left[2 \cos \left(\frac{\phi}{2} \right) \sin \left(\frac{2(kx - \omega t - \phi)}{2} \right) \right]$$

Hence, -

$$y = 2A_0 \cos \left(\frac{\phi}{2} \right) \sin \left(kx - \omega t - \frac{\phi}{2} \right)$$

this is the eqn of resultant wave.
where, -

$$2A \cos\left(\frac{\Phi}{2}\right) \Rightarrow \text{Resultant Amplitude with phase } \frac{\Phi}{2}$$

Condition:-

If $\Phi = 0$, $\cos\frac{\Phi}{2} = 1$; resultant amplitude $= 2A_0$ { constructive interference }

* Interference of Waves:-

When two waves of same frequency, same speed moving in the same ~~speed~~ direction in same medium superimpose, such effect of superposition of waves is called Interference of waves.

Types:-

① Constructive:-

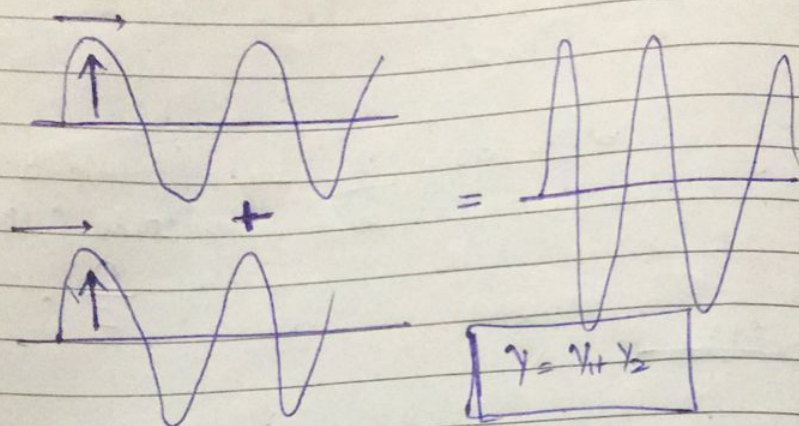
Interference of two or more waves

in such a way that crest of one wave superimpose with crest of another wave, such interference of waves is called "constructive interference".

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Q:- Diff. between beats & frequency

Date _____



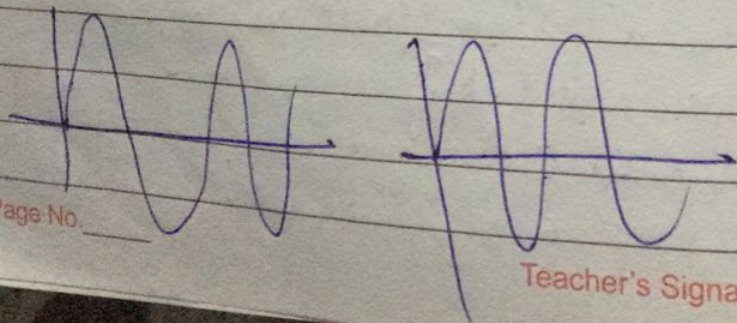
Condition for ~~card~~ constructive interference:-

$$\text{Path Difference} = m\lambda$$

$$\text{where } m = 0, 1, 2, 3, \dots, n$$

② Destructive Interference

Interference of two or more waves in such a way that crest of one wave superimpose with trough of another wave, such interference of waves is called "Destructive Interference".



Ques Data:-

wave equation :- $y(x,t) = 8.0 \sin \left[\frac{\pi}{3.0} x - 66t \right] \text{ cm}$

Amplitude $= A_0 = ?$
 $\lambda = ?$

$$y \text{ of } x, t = A \sin(kx - \omega t) \quad \text{--- (1)}$$

compare eq (1) & (2) :-

a) $A_0 = 3.5 \text{ cm}$

b) $\therefore k = \frac{2\pi}{\lambda}$

from eq (1)

$$k = \frac{\pi}{3}$$

$$\frac{2\pi}{\lambda} = \frac{\pi}{3}$$

$$\boxed{6 = \lambda \text{ m.}}$$

(Additional):-

for v :-

$$v = \frac{\omega}{k} = \frac{\text{coeff. of } t}{\text{coeff. of } x} = \frac{66}{\pi/3} = 70.0 \text{ cm/s}$$

Teacher's Signature

Qc Data:-

$$L = 10.0 \text{ m}$$

$$W_1 = 0.25 \text{ N (for string)}$$

$$W_2 = 1000 \text{ N} = 1000 \text{ N (for weight attached)}$$

$$t = ?$$

Solution :-

$$\therefore v = \sqrt{\frac{T}{\mu}}$$

$$\mu = \frac{m}{L}$$

for m

$$W = mg$$

$$m = \frac{0.25}{9.8}$$

$$= 0.0255 \text{ kg}$$

$$\mu = \frac{m}{L}$$

$$= \frac{0.0255}{10}$$

$$= 2.55 \times 10^{-3} \text{ kg/m}$$

$$\Rightarrow v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1000}{2.55 \times 10^{-3}}}$$

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$$V = 626.44 \text{ m/s}$$

$$V = \frac{s}{t}$$

$$t = \frac{s}{V}$$

$$t = \frac{L}{V}$$

$$= \frac{10}{626.44}$$

$$t = 0.01596 \text{ s}$$

Q6:- Data:-

$$A_0 = 5.0 \text{ cm}$$

$$A_n = 6.69 \text{ cm}$$

$$\phi = ?$$

$$y = 2A_0 \cos\left(\frac{\phi}{2}\right) \sin(kx - \omega t - \phi)$$

resultant amplitude.

Sol:-

$$A_r = 2A_0 \cos\left(\frac{\phi}{2}\right)$$

$$\frac{A_R}{2 \times A_0} = \cos\left(\frac{\phi}{2}\right)$$

$$\frac{A_R}{2A} = \cos\left(\frac{\phi}{2}\right)$$

$$0.669 = \cos\left(\frac{\phi}{2}\right)$$

$$\cos^{-1}(0.669) = \frac{\phi}{2}$$

$$48.01 = \frac{\phi}{2}$$

$$48.01 \times 2 = \phi$$

$$\boxed{\phi = 96.0^\circ}$$

Q7:-

Let :-

$\Delta f_1 \rightarrow$ fundamental frequency
 $= -4.0\% \text{ of } f_1$

$$f' - f_1 = -\frac{4}{100} f_1$$

$$\Delta f_1 = f'_1 - f_1$$

$$f'_1 = f_1 - 0.04 f_1$$

$$\boxed{f'_1 = 0.96 f_1}$$

Now:-

$$\Delta T = 2\% \text{ age}$$

Sol:-

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{4}} \quad \text{--- (1) Initial state.}$$

$$f_1' = \frac{1}{2L} \sqrt{\frac{T'}{4}} \quad \text{--- (2) Final state.}$$

$$\therefore f' \propto T'$$

Divide eq (1) by (2)

$$\frac{f_1}{f_1'} = \sqrt{\frac{T}{T'}}$$

S.B.S

$$\left(\frac{f_1}{f_1'} \right)^2 = \frac{T}{T'}$$

$$\left(\frac{f_1}{0.96 f_1'} \right)^2 = \frac{T}{T'}$$

$$\left(\frac{1}{0.96} \right)^2 = \frac{T}{T'}$$

$$\left(\frac{1}{0.9216} \right) = \frac{T}{T'}$$

$$T' = 0.9216 T$$

$$\Delta T = T' - T$$

$$= 0.9216 T - T$$

$$\Delta T = -0.0784 T$$

In form of %age

$$\Delta T = -7.84\% \text{ of } T$$

Hence,

The period reduces by 7.84%.

Q9 → ~~Rece~~ Doppler derivation.

Q10:- Data:-

$$f' = 219 \text{ Hz (Approach)}$$

$$f = 184 \text{ Hz (Receding)}$$

↓
Stationary.

$$V = 340 \text{ m/s}$$

$$V_{\text{source}}/V_{\text{train}} = ?$$

$$f = ? \text{ (Actual)}$$

Sol:-

In case of approaching:-

$$f' = \left(\frac{V}{V - V_s} \right) f$$

$$219 = \left(\frac{V}{V - V_s} \right) f \quad \text{--- (1)}$$

Q2, Q3 → Reasoning

Q4 → Calculation

$$\Delta P = 100 \text{ k Pascal} \rightarrow 10^5 \text{ Pascal}$$

$$\frac{\Delta V}{V} = 5 \times 10^{-2} \% = \frac{5 \times 10^{-3}}{100} \rightarrow 5 \times 10^{-5}$$

a) Bulk Modulus of Water = ?

b) Speed of Sound = ?

Formula:-

$$a) B = \frac{\Delta P}{\Delta V/V}$$

$$= \frac{10^5}{5 \times 10^{-5}}$$

$$B = 2 \times 10^9 \text{ Pa}$$

$$b) V = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{2 \times 10^9}{1000}}$$

$$V = 1414.21 \text{ m/s}$$

Diff. between travelling
 & stationary waves.
 Date

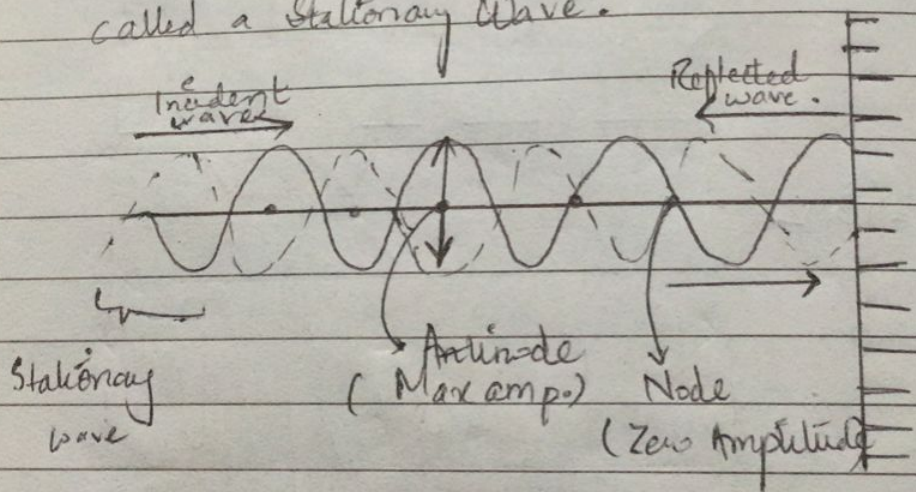
$$f_B = f_A - f_B$$

Unit = $\frac{\text{beats}}{\text{second}}$

* Stationary Waves / Standing Waves

Defn

When 2 plane waves having same amplitude and frequency travelling with same speed in same medium but in opposite direction along a line are super-imposed, a wave obtained is called a Stationary Wave.



• Reflected wave $\rightarrow$ out of phase. Crest & Trough get interchanged. $\rightarrow$
by π or 180°

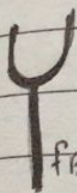
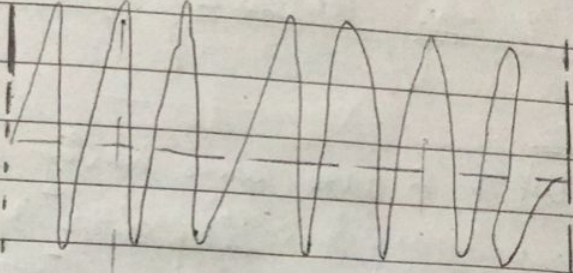
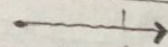
* BEATS :- (For transverse waves)

Def:-

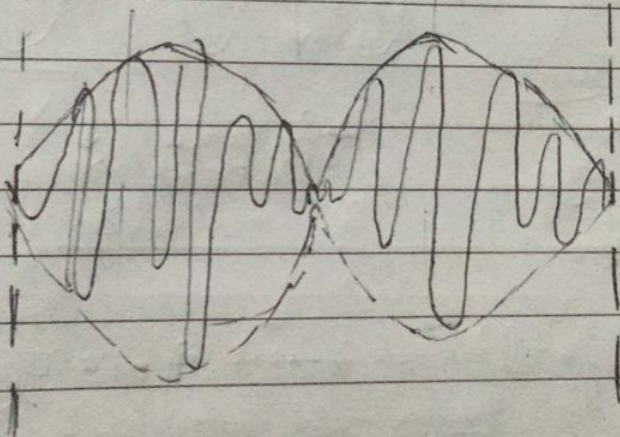
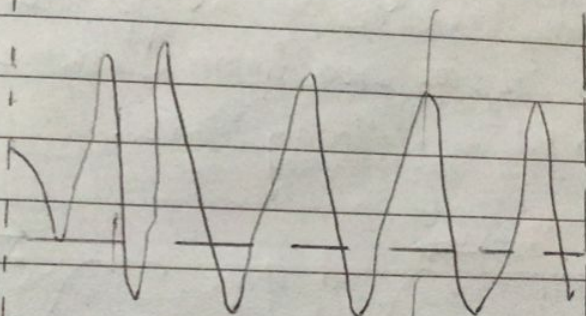
Beats are the effect of superposition of two waves having equal amplitude but slightly different frequencies moving with same speed in the same direction in the same medium.



$f_A = 15 \text{ Hz}$



$f_B = 13 \text{ Hz}$



- Beats - Periodic variation in amplitude of sound due to equal amplitude but diff. frequencies.
- Beat frequency f_B = No. of beats per second.

* Mathematical Treatment for Stationary Waves

wave equation is

Incident wave is

$$Y_1 = A \sin(kx - \omega t)$$

Reflected wave is

$$Y_2 = A \sin(kx + \omega t)$$

Resultant wave is

$$Y = Y_1 + Y_2$$

$$\begin{aligned} \Rightarrow Y &= A \sin(kx - \omega t) + A \sin(kx + \omega t) \\ &= A \left[\underbrace{\sin(kx - \omega t)}_{\alpha} + \underbrace{\sin(kx + \omega t)}_{\beta} \right] \end{aligned}$$

$$\left[\sin \alpha + \sin \beta = 2 \cos \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right) \right]$$

$$Y = A \left[2 \cos \left(\frac{kx - \omega t - (kx + \omega t)}{2} \right) \sin \left(\frac{kx - \omega t + kx + \omega t}{2} \right) \right]$$

$$Y = A \left[2 \cos \left(\frac{kx - \omega t - kx - \omega t}{2} \right) \sin \left(\frac{2kx}{2} \right) \right]$$

$$Y = A \left[2 \cos \left(\frac{-2\omega t}{2} \right) \sin(kx) \right]$$

$$y = A [2 \cos(-\omega t) \sin(kx)]$$

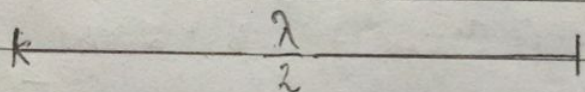
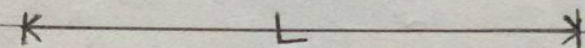
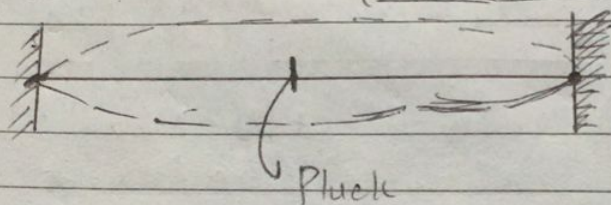
$$\therefore \cos(-0) = \cos 0$$

$$y = 2A \sin(kx) \cos(\omega t)$$

* Stationary Wave in Stretched

Spring - Modes of Vibration

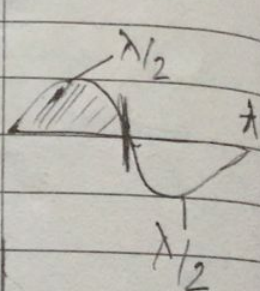
Mode: 01 (1st Harmonics)



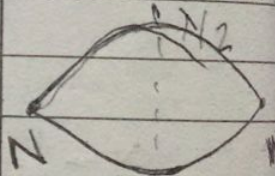
$$v = f \lambda$$

$$f = \frac{v}{\lambda}$$

$$f_1 = \frac{v}{\lambda_1}$$



Distance b/w 2 nodes



① Pluck at $\frac{\lambda}{2}$

② Pluck at $\frac{\lambda}{4}$

Where is Pluck?

Teacher's Signature

2nd - no. of

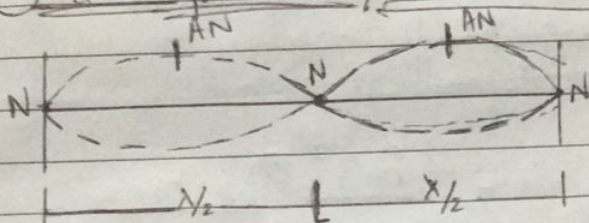
$$f_1 = \frac{v}{2L}$$

Speed of transverse waves is given by $v = \sqrt{\frac{T}{\mu}}$

$$f_1 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

→ 2nd Modes of Vibration

② 2nd Harmonic / 1st Overtone :-



$$L = \lambda/2 + \lambda/2 = 2 \times \lambda/2$$

Here :-

$$L = 2 \times \lambda/2$$

$$\therefore \lambda_2 = \frac{2L}{2}$$

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_2 = \frac{v}{\lambda_2}$$

Stationary
wave in
air

longitudinal.

* Organ Pipe :-

Date _____

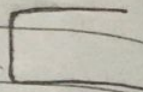
An organ pipe is a sound producing element of the pipe that resonates at a fixed pitch when pressurized air is driven through it.

e.g. - Flute.

Types :-

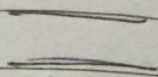
① Closed Organ Pipe :-

• One side is open & one is closed.



② Open Organ Pipe :-

• Both ends of pipe are open.

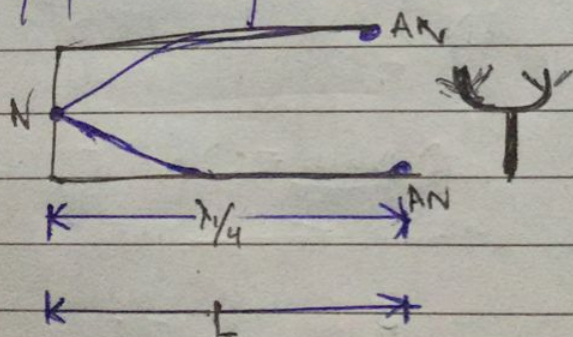


Stationary Waves In Organ Pipes

① Closed Organ Pipe :-

Modes of vibration :-

① 1st / Harmonic / Fundamental Tone :-



$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

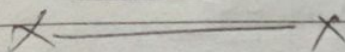
$$f_3 = \frac{v}{\lambda_3}$$

$$f_3 = \frac{v}{\frac{2L}{3}}$$

$$f_3 = 3 \left[\frac{v}{2L} \right]$$

$$\frac{v}{2L} = f_1$$

$$f_3 = 3f_1$$



Imp

No. of AN = No. of Loop

No. of Nodes = No. of Loops + 1.

Generalized Representation

(n) n^{th} Harmonic / $(n-1)^{\text{th}}$ Harmonic

$$f_n = n f_1$$

Teacher's Signature _____

$$\lambda = \frac{4L}{3}$$

$$\lambda_2 = \frac{4L}{3}$$

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_2 = \frac{v}{\lambda_2}$$

$$f_2 = \frac{v}{\frac{4L}{3}}$$

$$f_2 = 3\left(\frac{v}{4L}\right)$$

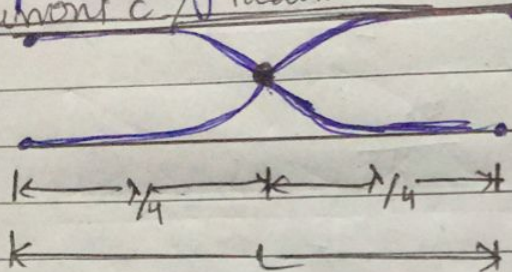
$$f_2 = 3f_1$$

Generalized expression:-

$$f_n = (2n-1)f$$

② Open Organ Pipe :-

1st Harmonic & Fundamental Tone:-



Here:-

$$L = 2 \times \frac{\lambda}{4}$$

$$\lambda_1 = \frac{4L}{2}$$

Now:-

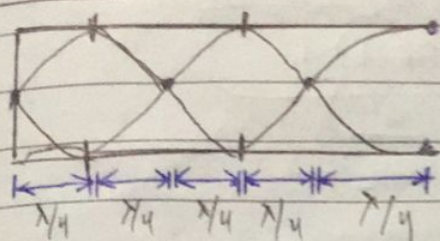
$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_1 = \frac{v}{\lambda_1}$$

Teacher's Signature

Q3) 5th Harmonic and Overtones



$$L = \frac{5\lambda}{4}$$

$$L$$

$$L = \frac{5\lambda}{4}$$

$$\lambda = \frac{4L}{5}$$

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_3 = \frac{v}{\lambda_3}$$

$$f_3 = \frac{v}{\frac{4L}{5}}$$

$$f_3 = 5 \left(\frac{v}{4L} \right) \rightarrow \text{Answer}$$

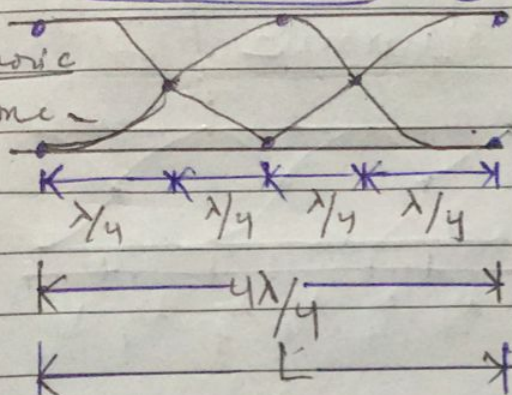
$$f_3 = 5f_1$$

$$f_1 = \frac{v}{\frac{4L}{2}}$$

$$f_1 = 2 \left(\frac{v}{4L} \right)$$

$$f_1 = \frac{v}{2L}$$

2nd Harmonic
1st overtone



$$L = \frac{4\lambda_2}{4}$$

$$\lambda_2 = \frac{4L}{4}$$

Now:-

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_2 = \frac{v}{\lambda_2}$$

$$f = \frac{v}{\frac{4L}{4}}$$

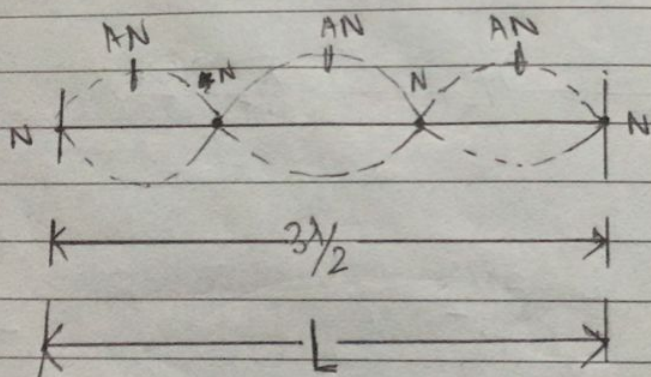
$$f_2 = \frac{v}{\frac{2L}{2}}$$

$$\Rightarrow f_2 = 2 \left[\frac{v}{2L} \right]$$

$$\frac{v}{2L} = f_1$$

$$\boxed{f_2 = 2f_1}$$

$\Rightarrow$ 3rd Harmonic / 2nd overtone



Here ;

$$L = 3 \times \frac{\lambda}{2}$$

$$\boxed{\lambda_3 = \frac{2L}{3}}$$

Here:-

$$L = \lambda/4$$

90°

$$\boxed{\lambda_1 = 4L}$$

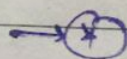
Now:-

$$v = f\lambda$$

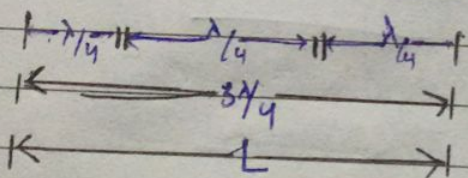
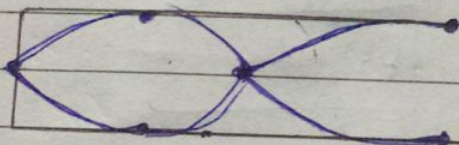
$$f = \frac{v}{\lambda}$$

$$f_1 = \frac{v}{\lambda_1}$$

$$\boxed{f_1 = \frac{v}{4L}}$$



② 2nd Harmonic / 1st Overtone

Here:-

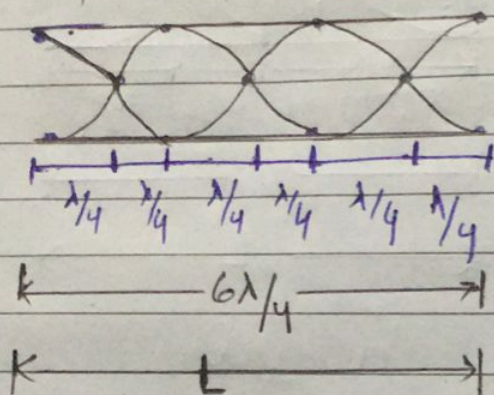
$$L = 3\lambda/4$$

$$f_2 = 4 \left(\frac{v}{4L} \right)$$

$$f_2 = 2 \left(\frac{v}{2L} \right)$$

$$f_2 = 2f_1$$

(iii) 3rd Harmonic / 2nd Overtone.



Here

$$L = \frac{6\lambda_3}{4}$$

$$\frac{4L}{6} = \lambda_3$$

Now:-

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$f_3 = \frac{v}{\lambda_3}$$

No. _____

Teacher's Signature _____

$$f_3 = \frac{v}{\frac{4L}{6}}$$

$$f_3 = 3 \left(\frac{v}{\frac{4L}{2}} \right)$$

$$f_3 = 3 \left(\frac{v}{2L} \right)$$

$$\boxed{f_3 = 3f_1} \rightarrow \textcircled{+}$$

General Terms

$$\boxed{f_n = nf_1}$$